

Infer-bench datasets

HaluEval, SQuAD, HotpotQA, TruthfulQA

300-example exports, citations, and shipped OnlineRFParm numbers

CFPerm / OnlineRFParm landing note

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Abstract

Four public NLP benchmarks are frozen as $n = 300$ local JSONL slices for inference-side landing work: HaluEval (Li et al., 2023), SQuAD (Rajpurkar et al., 2016), HotpotQA (Yang et al., 2018), and TruthfulQA (Lin et al., 2022). This note posts the slice inventory, characterization of those 300-row exports, and the OnlineRFParm numbers already run on HaluEval QA (plus HH-RLHF (Bai et al., 2022) as the alignment counterpart, not a fifth export). These tools detect / localize / rank under shift; they are not fact checkers.

1 Scope

The four exports live at `data/hf_cache/infer_bench_export/*.jsonl`. BibTeX is in the same folder (`infer_bench.bib` comprehensive; `recommended.bib` venue keys only). Related follow-ups kept in the comprehensive file: SQuAD 2.0 (Rajpurkar et al., 2018) and HaluEval 2.0 (Li et al., 2024). HaluEval QA is itself derived in part from HotpotQA, so both citations are appropriate when using that slice.

Table 1: Infer-bench inventory. Local slices are $n = 300$ head/mid/tail windows of the named Hugging Face split. Venue cite keys are in the last column.

Dataset	Task	Hub split	Source n	Export n	Cite key
HaluEval	Hallucination (QA)	<code>qa_samples</code>	10000	300	<code>li-etal-2023-halueval</code>
SQuAD 1.1	Extractive RC	validation	10570	300	<code>rajpurkar-etal-2016-squad</code>
HotpotQA	Multi-hop QA	distractor val.	7405	300	<code>yang-etal-2018-hotpotqa</code>
TruthfulQA	Truthfulness	generation val.	817	300	<code>lin-etal-2022-truthfulqa</code>

Table 2: Characterization of the $n = 300$ local JSONL exports. SQuAD validation is article-grouped, so three windows cover few titles.

Dataset	Q words	Ans. words	Extra	Notes
HaluEval	17.3	6.7	halluc. yes 51.3%	knowledge 55 words
SQuAD	9.6	2.0	3 titles	context 106 words
HotpotQA	15.6	2.4	bridge 234 / cmp 66	all hard; ≈ 10 paras
TruthfulQA	10.6	9.1	adv. 179 / non 121	27 categories

2 What is not claimed

`po_risk0` ranks audit candidates after a regime fire. It is residual strength versus the previous probe, not a factuality score and not unique causal attribution to a single feature.

Table 3: OnlineRFPPerm landing results already shipped on the HF-prototype branch. HaluEval uses a 2400-row `qa.samples` stream (not the 300-export). HH-RLHF is the alignment counterpart, not one of the four infer-bench exports.

Stream	n	Cut t	Fire rate	First fire	Pre→post	Ranking
HaluEval QA	2400	B_4	0.091	4	halluc. 0.080 → 0.942	AUROC <code>po_risk0</code> = 1.00
HH-RLHF helpful	4000	B_4	0.104	4	judge err 0.147 → 0.539	style AUC 0.999

Table 4: HaluEval OnlineRFPPerm `hop@cut` vs. a quiet batch. Gate $\gamma = 1.25$. `po_risk0` ranks audit candidates; it is not a fact checker.

Batch	Fired	Ratio	P@10	Halluc. rate
Quiet $t = 2$	no	0.70	0.70	0.07
Cut $t = 4$	yes	11.71	1.00	0.82

References

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